ARSHIA NAEEM

Lahore, Pakistan | arshya.naeem@gmail.com | LinkedIn | GitHub | Portfolio

EDUCATION

Master of Science in Computer Science
Lahore College for Women University, Lahore, Pakistan | 2015 – 2017

- **Academic Standing:** CGPA 3.74 / 4.00 (3rd Position & Roll of Honor, MS-CS Batch)
- **Thesis Topic:** Agglomerative Hierarchical Clustering Optimization (Published in *Current Science*, 2019)
- **Core Specialization:** Unsupervised Machine Learning, Advanced Algorithm Analysis, Deep Learning, Statistical Data Analysis

Bachelor of Science (Hons.) in Computer Science
Lahore College for Women University, Lahore, Pakistan | 2011 – 2015

- **Academic Standing:** CGPA 3.62 / 4.00
- **Final Year Project:** AXPERT: A Web-Based Agriculture Information System (Published in *PJAR*, 2016)
- **Core Electives:** Artificial Intelligence, Software Engineering, Database Systems, Digital Image Processing

PROFESSIONAL EXPERIENCE

Founder & Technical Lead — Asan Kasht Kari
K-BIC Incubation Center, Lahore College for Women University | **2025 – 2026**

- Designed and built the ML pipeline behind an AI-driven agricultural optimization platform from concept to incubation graduation (Cohort-II, K-BIC, LCWU). The farmer-facing app records crop expenses and income by task and season, reports profit and analytics, and detects crop disease.

Senior Lecturer (Computer Science) **Sep 2024 – Present**
Department of Computer Science, University of Central Punjab (UCP), Lahore, Pakistan

- Taught core CS courses (Compiler Construction, Databases, Software Engineering, Operating Systems) to 120+ students/semester using custom Project-Based Learning frameworks; advise on infrastructure specifications for new Edge AI and Robotics labs.

Senior Lecturer & Quality Circle Head (QCH) **Feb 2022 – Sep 2024**
Department of CS & IT, Superior University, Lahore, Pakistan

- Supervised 5–10 applied student R&D projects annually (algorithm-to-prototype translation, 300+ students/year) while leading CLO quality-assurance mapping for the department.

Entrepreneurial Instructor & Venture Mentor **Aug 2021 – Sep 2024**
Chaudhary Muhammad Akram Center for Entrepreneurship & Development (CMACED), Superior University

- Mentored 24 student ventures from ideation to execution across Agritech, Climate Tech, and Metaverse domains.

System Developer & Research Internee **May 2016 – Aug 2016**
The Eye Associates, Lahore, Pakistan

- Independently designed and built an online Electronic Medical Record (EMR) system (Java Servlets, Struts2, MySQL Workbench) supporting clinical checkups, surgery scheduling, and patient history management.

PUBLICATIONS & RESEARCH DISSEMINATION

Journal Articles

- Hameed, K., **Naeem, A.,** Maqsood, F., & Naha, R. (2026). "A Quantum-Resilient and Forensic-Aware Security Lifecycle Framework for AI-Driven Edge Sensing in Intelligent Healthcare IoT Systems."
- **Naeem, A.**, Rehman, M., Anjum, M., & Asif, M. (2019). "Development of an efficient hierarchical clustering analysis using an agglomerative clustering algorithm." Current Science, 117(6), 1045–1053.
- **Naeem, A.**, Anjum, M., Rehman, M., Mahmood, Z., & Kamran, M. A. (2016). "An Integrated Information System to Facilitate Farmers in Wheat, Sugarcane and Other Crop Diseases Identification." Pakistan Journal of Agricultural Research, 29(4).

Peer-Review Service

- Reviewer, IEEE Access (2019) — "Ultra-Fast and Accurate Multi-Touch Tracking Algorithm for a Large Number of Points."

Poster Presentations & Research Seminars

- Invited research seminar, "Agriculturist Facilitating System," Ayub Agriculture Research Institute, Faisalabad (June 2015).
- Poster presentation, "Agriculturist Facilitating System," 5th Invention to Innovation Summit, University of the Punjab, Lahore (March 2016).
- Poster presentation, "Agriculturist Facilitating System," 7th International Workshop on Field & Assistive Robotics, FAST-NUCES, Lahore (October 2014).

RESEARCH INTERESTS

- **Computer Vision for Agriculture:** crop disease detection and segmentation in real field conditions
- **Robust and Low-Resource Machine Learning:** distribution shift, small data, class imbalance, model interpretability
- **Human-Centered, Farmer-Facing AI:** simple decision-support tools, human-in-the-loop systems
- **Applied Computer Vision:** medical imaging as a second application area

ONGOING RESEARCH AND PROJECTS

Systematic Mapping Study on Usability of Mobile Applications in E-Agriculture | 2023–Present
Conducting a systematic mapping study analyzing publication trends, mobile application domains, software architectures, and AI/IoT technique usability across agricultural sectors. Currently screening and extracting studies from 350+ primary sources retrieved from major databases, en route to identifying key technological gaps, usability barriers for rural end-users, and performance metrics. Manuscript targeted for journal submission upon completion of extraction and analysis.

Adaptive Learning and Emotional Intelligence Platform for Early Childhood Education | 2026–Present

Designed an AI-driven adaptive learning platform combining Random Forest classification and Item Response Theory (IRT), grounded in Vygotsky's Zone of Proximal Development and Bloom's Taxonomy. Implementation is currently underway, targeting platform deployment and manuscript submission by late 2026. Department of Computer Science, University of Central Punjab, Lahore, Pakistan.

RESEARCH & PROJECT LEADERSHIP

- **AI-Based Flood Monitoring and Warning System (2023)**: Personally chose the modeling approach and conducted the remote-sensing data analysis for a real-time flood-forecasting system, while overseeing student implementation of the regression/time-series pipeline (Python).
- **Diagnosis of Conjunctivitis / Pink Eye using AI (2024)**: Supervised a student-built computer-vision pipeline (Python, NumPy, CNNs) classifying healthy vs. infected eyes, including guidance on addressing dataset class imbalance via SMOTE.
- **ETMS: Effortless Traffic Management System (2024)**: Supervised an AI-powered urban traffic-control system using OpenCV for real-time camera processing and predictive ML for signal-timing optimization; funded/shortlisted by IGNITE NGIRI.
- **Virtual Influencer (2024)**: Supervised development of Metaverse avatars with independent decision-making using computer vision, generative AI, and reinforcement learning; awarded a national R&D grant.
- **Sagawere (2022)**: Supervised a gesture-controlled VR interface using Mediapipe computer vision; awarded, National Startup Championship.

GRANTS, COMPETITIONS & STARTUP SUPPORT

- **Founder & Technical Lead:** Asan Kasht Kari | K-BIC Incubation Center, LCWU (2025–2026) — Successfully incubated and graduated, Cohort-II.
- **Project Supervisor:** Effortless Traffic Management System (ETMS) | IGNITE NGIRI R&D Grant (national competitive program), 2023–2024 — Shortlisted.
- **Project Supervisor:** Dynamic Sun Tracking Solution | IGNITE NGIRI R&D Grant (national competitive program), 2023–2024 — Shortlisted.
- **Startup Instructor:** Virtual Influencer | SEE Pakistan 2024 Grant — Awarded, national R&D grant (~PKR 244,000).
- **Project Supervisor:** Locomotion of Gaming | IGNITE NGIRI R&D Grant (national competitive program), 2021–2022 — Awarded (~PKR 100,000).
- **Project Supervisor:** Ever-Tracker | IGNITE NGIRI R&D Grant (national competitive program), 2021–2022 — Awarded (~PKR 100,000).
- **Project Supervisor**: SAGAWERE | National Startup Championship, 2022 — Awarded, national R&D grant (~PKR 12,200).

AWARDS & HONORS

- **Graduation Certificate – Cohort-II:** Khadija-tul-Kubra Business Incubation Center (K-BIC), LCWU, for successfully founding and graduating Agritech startup *Asan Kasht Kari* (2026).
- **Rector's Performance of Excellence Award:** Honored as "Outstanding Performer of the Year" in CS & IT Dept, Superior University (2021).
- **Academic Excellence & Roll of Honor:** Secured 3rd Position overall in MS-CS batch (CGPA 3.74) at LCWU 14th Convocation (2017).

CERTIFICATIONS & ADVANCED TECHNICAL TRAINING

- **Specialized Research Training:** Intensive Workshop: *Computer Vision, Machine Learning, and its Applications through Edge Devices* – NCRA-Agriculture Robotics Lab, **LUMS**, Lahore (2022).
- Workshop on Microsoft Azure Cloud Platform from 6th-8th January 2015.
- **Coursera Professional Certifications:** *Introduction to Systematic Review and Meta-Analysis* (Data Aggregation & Statistical Verification). *Programming for Everybody:* (Getting Started with Python), Python Data Structures, Using Python to Access Web Data, *C for Everyone:* Programming Fundamentals & Structured Programming, *Cracking the Creativity Code:* Discovering Ideas.

TECHNICAL SKILLS

- **Languages:** Python (primary), C++, C#, MATLAB, SQL, Java
- **Deep Learning & Computer Vision:** PyTorch, TensorFlow/Keras, OpenCV; CNNs, YOLOv8, UNet; object detection and segmentation
- **Data Analysis & Visualization:** NumPy, Pandas, Scikit-Learn, Matplotlib, Jupyter
- **Tools & Deployment:** Git/GitHub, Docker, FastAPI, Flask, Linux
- **Research Tools:** LaTeX, Zotero/Mendeley, SPSS

LANGUAGE SKILLS

English (Fluent), Urdu (Native), Punjabi (Native)

REFERENCES

Available upon request